Exercise 1 – Happy Supplies Parts Order Form

a. Assumptions

1. Each form is one order. The form has no order number and surrogate keys are not allowed, so an order is identified by (customerNumber, orderDate, orderTime). A customer does not place two orders at the same date and time.

2. customerNumber uniquely identifies a customer and determines customerName and customerType.

3. Each order is handled by one employee. Employees can help any customer, so the employee depends on the order, not on the customer. The employee is only recorded by name, and there are no other employee attributes, so no separate employee table is created.

4. partNumber uniquely identifies a part and determines partName, partType and cageCode. Each part is stored in one cage.

5. unitPrice is the standard price of the part. It is the same for every customer and order, so it depends on partNumber.

6. A part appears at most once in an order. If more units are needed, quantityOrdered is increased.

7. cageCode and customerType have no other attributes on the form, so they do not get their own tables. A cage can hold parts of different types and prices, so cageCode does not determine any other attribute.

b. Original data (unnormalized)

partOrder

customerNumber	customerName	customerType	**orderDate**	**orderTime**	employeeName	partNumber	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654 10456 10776 10657	Float Control Modulator Hose Assembly Float Assembly	Plumbing Electrical Plumbing Plumbing	G413 H433 G413 G413	4 3 7 5	12 7 9 10

(customerNumber, orderDate, orderTime) identifies the form, but it cannot identify a single value in the part columns, because each part column holds four values. This is a repeating group.

The table is not in any normal form.

First Normal Form (1NF)

partOrder – 1NF

customerNumber	customerName	customerType	**orderDate**	**orderTime**	employeeName	**partNumber**	partName	partType	cageCode	quantityOrdered	unitPrice
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10654	Float Control	Plumbing	G413	4	12
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10456	Modulator	Electrical	H433	3	7
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10776	Hose Assembly	Plumbing	G413	7	9
HG54587	Jeff Peterson	Consumer	7/1/2024	10:30 am	D. Harrison	10657	Float Assembly	Plumbing	G413	5	10

The repeating group is removed by putting each ordered part on its own row. The order and customer values are repeated on each row.

Each cell now has one atomic value, so the table is in 1NF.
Since the same order now appears on four rows, partNumber has to be added to the key.
PK = (customerNumber, orderDate, orderTime, partNumber).
Functional dependencies:
customerNumber → customerName, customerType (partial dependency)
(customerNumber, orderDate, orderTime) → employeeName (partial dependency)
partNumber → partName, partType, cageCode, unitPrice (partial dependency)
(customerNumber, orderDate, orderTime, partNumber) → quantityOrdered (full dependency)
The table is not in 2NF because several attributes depend on only part of the composite key.

Second Normal Form (2NF)

customer – 2NF

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

customerName and customerType depend only on customerNumber, which is part of the composite key. They are moved into customer. PK = customerNumber.

customerOrder – 2NF

customerNumber	**orderDate**	**orderTime**	employeeName
HG54587	7/1/2024	10:30 am	D. Harrison

employeeName depends only on (customerNumber, orderDate, orderTime), not on partNumber. It is moved into customerOrder. PK = (customerNumber, orderDate, orderTime). customerName and customerType cannot stay here, because they depend only on customerNumber, which would still be a partial dependency.

part – 2NF

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

partName, partType, cageCode and unitPrice depend only on partNumber. They are moved into part. PK = partNumber.

orderLine – 2NF

customerNumber	**orderDate**	**orderTime**	**partNumber**	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

quantityOrdered depends on the whole key, so it stays in orderLine with the full composite key. PK = (customerNumber, orderDate, orderTime, partNumber).

Third Normal Form (3NF)

Each 2NF table is checked for transitive dependencies (a non-key attribute that depends on another non-key attribute):

customer: customerName and customerType both depend directly on customerNumber. There is no transitive dependency.

customerOrder: employeeName is the only non-key attribute, so there cannot be a transitive dependency.

part: partName, partType, cageCode and unitPrice all describe the part directly. By assumption 7, cageCode does not determine partType or unitPrice, so there is no transitive dependency.

orderLine: quantityOrdered is the only non-key attribute, so there cannot be a transitive dependency.

The 2NF tables are therefore already in 3NF. The data does not change. Foreign keys are shown in italics below.

customer – 3NF

customerNumber	customerName	customerType
HG54587	Jeff Peterson	Consumer

customerOrder – 3NF

customerNumber	**orderDate**	**orderTime**	employeeName
HG54587	7/1/2024	10:30 am	D. Harrison

part – 3NF

partNumber	partName	partType	cageCode	unitPrice
10654	Float Control	Plumbing	G413	12
10456	Modulator	Electrical	H433	7
10776	Hose Assembly	Plumbing	G413	9
10657	Float Assembly	Plumbing	G413	10

orderLine – 3NF

customerNumber	***orderDate***	***orderTime***	***partNumber***	quantityOrdered
HG54587	7/1/2024	10:30 am	10654	4
HG54587	7/1/2024	10:30 am	10456	3
HG54587	7/1/2024	10:30 am	10776	7
HG54587	7/1/2024	10:30 am	10657	5

c. Primary keys and foreign keys (3NF)

customer: PK = customerNumber

customerOrder: PK = (customerNumber, orderDate, orderTime); FK = customerNumber → customer.customerNumber

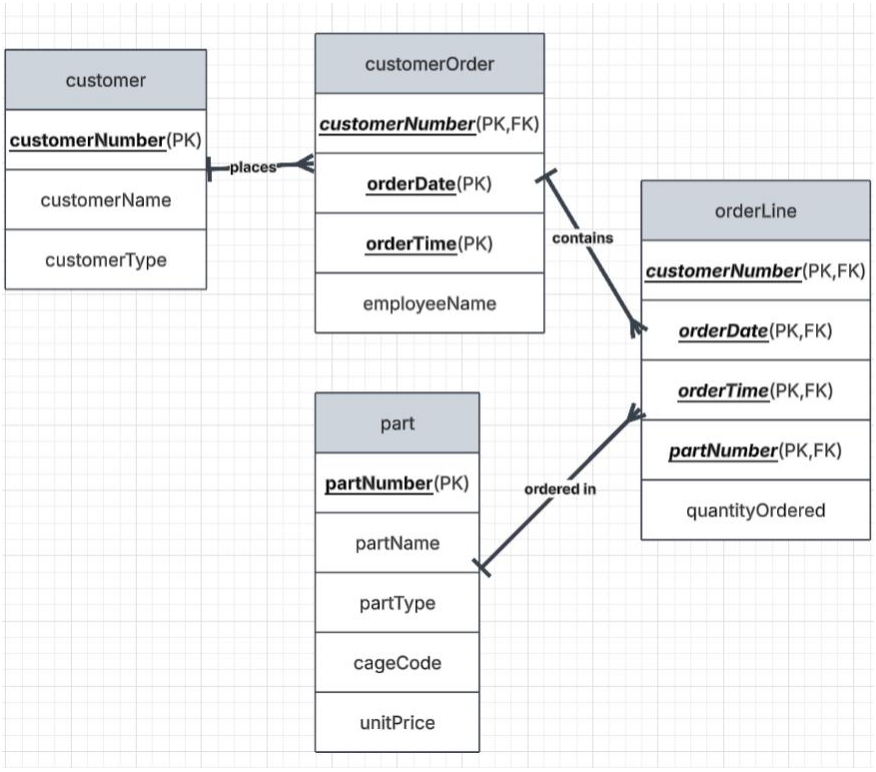
part: PK = partNumber

orderLine: PK = (customerNumber, orderDate, orderTime, partNumber); FK = (customerNumber, orderDate, orderTime) → customerOrder; FK = partNumber → part.partNumber

d. 3NF relations

customer, customerOrder, part, orderLine

ERD (3NF)


customer
customerNumber(PK)
customerName
customerType
places
customerOrder
customerNumber(PK,FK)
orderDate(PK)
orderTime(PK)
employeeName
contains
orderLine
customerNumber(PK,FK)
orderDate(PK,FK)
orderTime(PK,FK)
partNumber(PK,FK)
quantityOrdered
part
partNumber(PK)
partName
partType
cageCode
unitPrice
ordered in

Exercise 2 – Panacea Mental Health Corporation

a. Assumptions

1. staffNo uniquely identifies a therapist and determines therapistName.
2. patNo uniquely identifies a patient and determines patName.
3. A therapist can work at different branches on different days, but only at one branch on a given day. So (staffNo, appointmentDate) determines branchNo.
4. Each appointment has one therapist, one patient, one date, one time and one branch.
5. A therapist cannot have two appointments at the same date and time. A patient also cannot have two appointments at the same date and time.
6. The appointment column is stored as two attributes, appointmentDate and appointmentTime.
7. branchNo has no other attributes in the data, so no separate branch table is created.

b. Original data / First Normal Form (1NF)

appointment – 1NF

staffNo	therapistName	patNo	patName	**appointmentDate**	**appointmentTime**	branchNo
S1011	Fred Smith	P100	Lily White	9/12/2022	10:00	M15
S1011	Fred Smith	P105	Jill Baker	9/12/2022	12:00	M15
S1024	Heidi Pierce	P108	Andy McKee	9/12/2022	10:00	Q10
S1024	Heidi Pierce	P108	Andy McKee	9/14/2022	14:00	Q10
S1032	Richard Levin	P105	Jill Baker	9/14/2022	16:30	M15
S1032	Richard Levin	P110	Jimmy Winter	9/15/2022	18:00	B13

Every cell has one atomic value and there are no repeating groups, so the original table is already in 1NF.

Candidate keys are (staffNo, appointmentDate, appointmentTime) and (patNo, appointmentDate, appointmentTime). I chose (staffNo, appointmentDate, appointmentTime) as the PK.

Functional dependencies:

staffNo → therapistName (partial dependency)

(staffNo, appointmentDate) → branchNo (partial dependency)

(staffNo, appointmentDate, appointmentTime) → patNo (full dependency)

patNo → patName (transitive dependency)

The table is not in 2NF because therapistName and branchNo depend on only part of the key.

Second Normal Form (2NF)

therapist – 2NF

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapistName depends only on staffNo, which is part of the key. It is moved into therapist. PK = staffNo.

therapistSchedule – 2NF

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

branchNo depends only on (staffNo, appointmentDate), which is part of the key. It is moved into therapistSchedule. PK = (staffNo, appointmentDate). This table also records that a therapist is at one branch per day.

appointment – 2NF

staffNo	**appointmentDate**	**appointmentTime**	patNo	patName
S1011	9/12/2022	10:00	P100	Lily White
S1011	9/12/2022	12:00	P105	Jill Baker
S1024	9/12/2022	10:00	P108	Andy McKee
S1024	9/14/2022	14:00	P108	Andy McKee
S1032	9/14/2022	16:30	P105	Jill Baker
S1032	9/15/2022	18:00	P110	Jimmy Winter

patNo and patName stay in appointment because they do not depend on part of the key. PK = (staffNo, appointmentDate, appointmentTime).

This table is in 2NF but not in 3NF, because of the transitive dependency (staffNo, appointmentDate, appointmentTime) → patNo → patName. Andy McKee and Jill Baker are each stored twice.

Third Normal Form (3NF)

therapist – 3NF (unchanged)

staffNo	therapistName
S1011	Fred Smith
S1024	Heidi Pierce
S1032	Richard Levin

therapistName is the only non-key attribute. No transitive dependency.

therapistSchedule – 3NF (unchanged)

staffNo	**appointmentDate**	branchNo
S1011	9/12/2022	M15
S1024	9/12/2022	Q10
S1024	9/14/2022	Q10
S1032	9/14/2022	M15
S1032	9/15/2022	B13

branchNo is the only non-key attribute. No transitive dependency.

patient – 3NF (new)

patNo	patName
P100	Lily White
P105	Jill Baker
P108	Andy McKee
P110	Jimmy Winter

patName depends on patNo, not directly on the appointment key. It is moved into patient. PK = patNo.

appointment – 3NF (patName removed)

staffNo	***appointmentDate***	**appointmentTime**	*patNo*
S1011	9/12/2022	10:00	P100
S1011	9/12/2022	12:00	P105
S1024	9/12/2022	10:00	P108
S1024	9/14/2022	14:00	P108
S1032	9/14/2022	16:30	P105
S1032	9/15/2022	18:00	P110

patName is removed and patNo stays as a foreign key to patient. The appointment's branch is found through (staffNo, appointmentDate) in therapistSchedule.

c. Primary keys and foreign keys (3NF)

therapist: PK = staffNo

therapistSchedule: PK = (staffNo, appointmentDate); FK = staffNo → therapist.staffNo

patient: PK = patNo

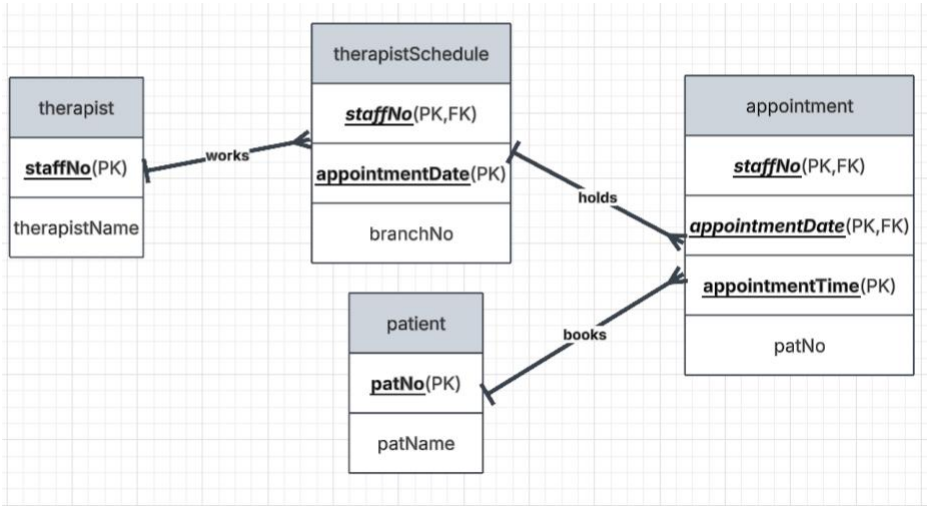
appointment: PK = (staffNo, appointmentDate, appointmentTime); FK = (staffNo, appointmentDate) → therapistSchedule; FK = patNo → patient.patNo

(patNo, appointmentDate, appointmentTime) is also unique in appointment, so a patient cannot be booked twice at the same time.

d. 3NF relations

therapist, therapistSchedule, patient, appointment

ERD (3NF)


therapist
staffNo(PK)
therapistName
works
therapistSchedule
staffNo(PK,FK)
appointmentDate(PK)
branchNo
holds
appointment
staffNo(PK,FK)
appointmentDate(PK,FK)
appointmentTime(PK)
patNo
patient
patNo(PK)
patName
books

Exercise 3 – Maid Better Temp Agency

a. Assumptions

1. eNo uniquely identifies an employee and determines eName. eName is not unique.
2. contractNo uniquely identifies a contract. Each contract applies to exactly one event.
3. An event can have several contracts for different service needs.
4. Each event has one location, so eventNo determines eventLoc.
5. hours is the total hours an employee worked on a contract, so there is one row for each (eNo, contractNo) pair.
6. An employee can work on several contracts, and a contract can have several employees.

b. Original data / First Normal Form (1NF)

employeeContract – 1NF

eNo	**contractNo**	hours	eName	eventNo	eventLoc
1135	C1024	16	Smith J	H25	Queens
1057	C1024	24	Hocine D	H25	Queens
1068	C1025	28	White T	H4	Yonkers
1135	C1025	15	Smith J	H4	Yonkers
1135	C1026	10	Smith J	H25	Queens

Every cell has one atomic value and there are no repeating groups, so the original table is already in 1NF.

Neither eNo nor contractNo is unique alone, but each (eNo, contractNo) pair appears once. PK = (eNo, contractNo).

Functional dependencies:

(eNo, contractNo) → hours (full dependency)

eNo → eName (partial dependency)

contractNo → eventNo (partial dependency)

eventNo → eventLoc (transitive dependency, so contractNo → eventNo → eventLoc)

The table is not in 2NF because eName, eventNo and eventLoc depend on only part of the key.

Second Normal Form (2NF)

employee – 2NF

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

eName depends only on eNo, which is part of the key. It is moved into employee. PK = eNo.

contract – 2NF

contractNo	eventNo	eventLoc
C1024	H25	Queens
C1025	H4	Yonkers
C1026	H25	Queens

eventNo and eventLoc depend only on contractNo, which is part of the key. They are moved into contract. PK = contractNo.

employeeContract – 2NF

eNo	**contractNo**	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

hours depends on the whole key, so it stays in employeeContract. PK = (eNo, contractNo).

The contract table is in 2NF but not 3NF, because contractNo → eventNo → eventLoc is a transitive dependency. H25 / Queens is stored twice (C1024 and C1026).

Third Normal Form (3NF)

employee – 3NF (unchanged)

eNo	eName
1135	Smith J
1057	Hocine D
1068	White T

eName is the only non-key attribute. No transitive dependency.

event – 3NF (new)

eventNo	eventLoc
H25	Queens
H4	Yonkers

eventLoc depends on eventNo, not directly on contractNo. It is moved into event. PK = eventNo.

contract – 3NF (eventLoc removed)

contractNo	*eventNo*
C1024	H25
C1025	H4
C1026	H25

eventLoc is removed and eventNo stays as a foreign key to event.

employeeContract – 3NF (unchanged)

eNo	***contractNo***	hours
1135	C1024	16
1057	C1024	24
1068	C1025	28
1135	C1025	15
1135	C1026	10

hours is the only non-key attribute and depends on the whole key. No transitive dependency.

c. Primary keys and foreign keys (3NF)

employee: PK = eNo

event: PK = eventNo

contract: PK = contractNo; FK = eventNo → event.eventNo

employeeContract: PK = (eNo, contractNo); FK = eNo → employee.eNo; FK = contractNo → contract.contractNo

d. 3NF relations

employee, event, contract, employeeContract

ERD (3NF)

